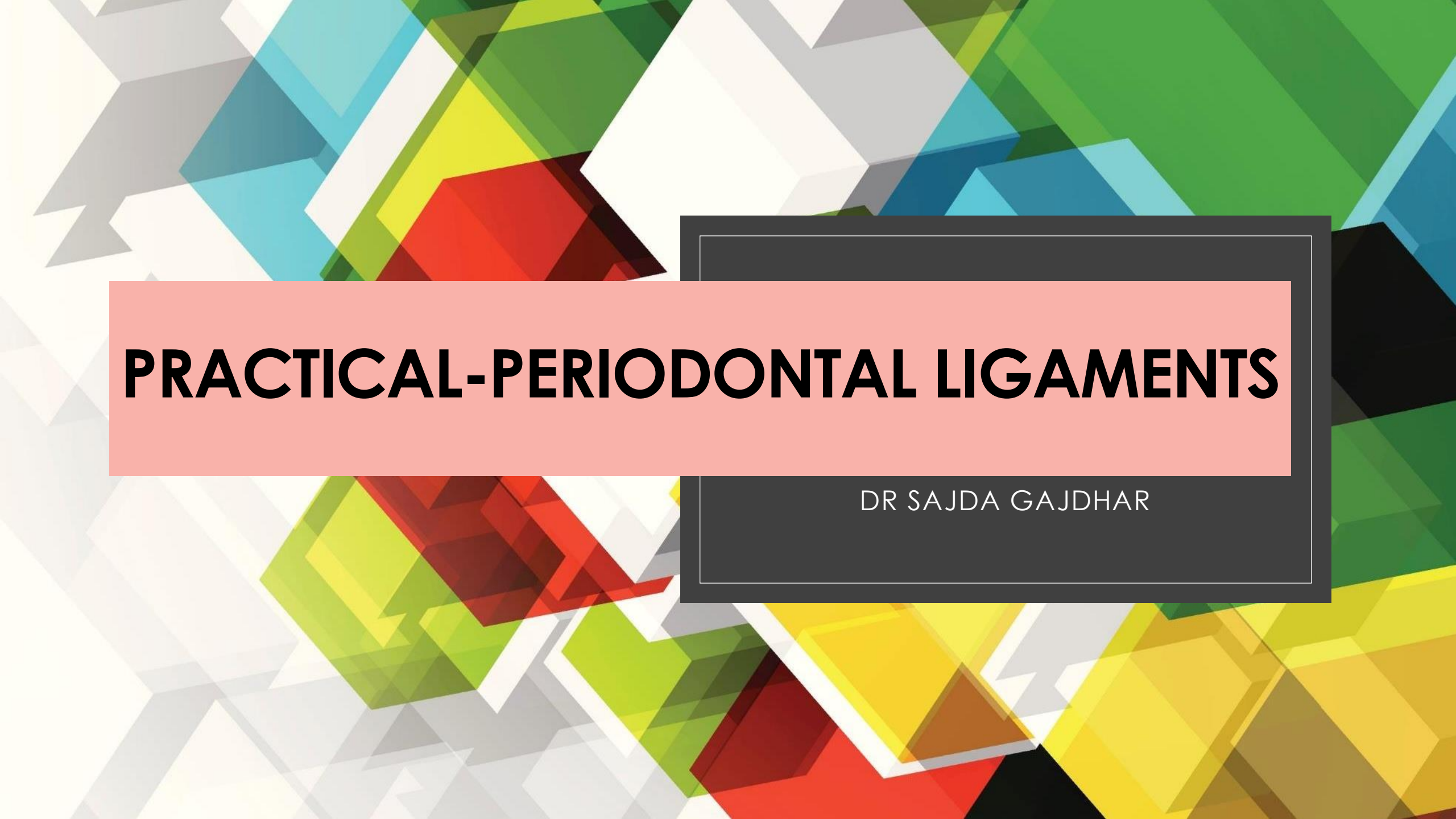


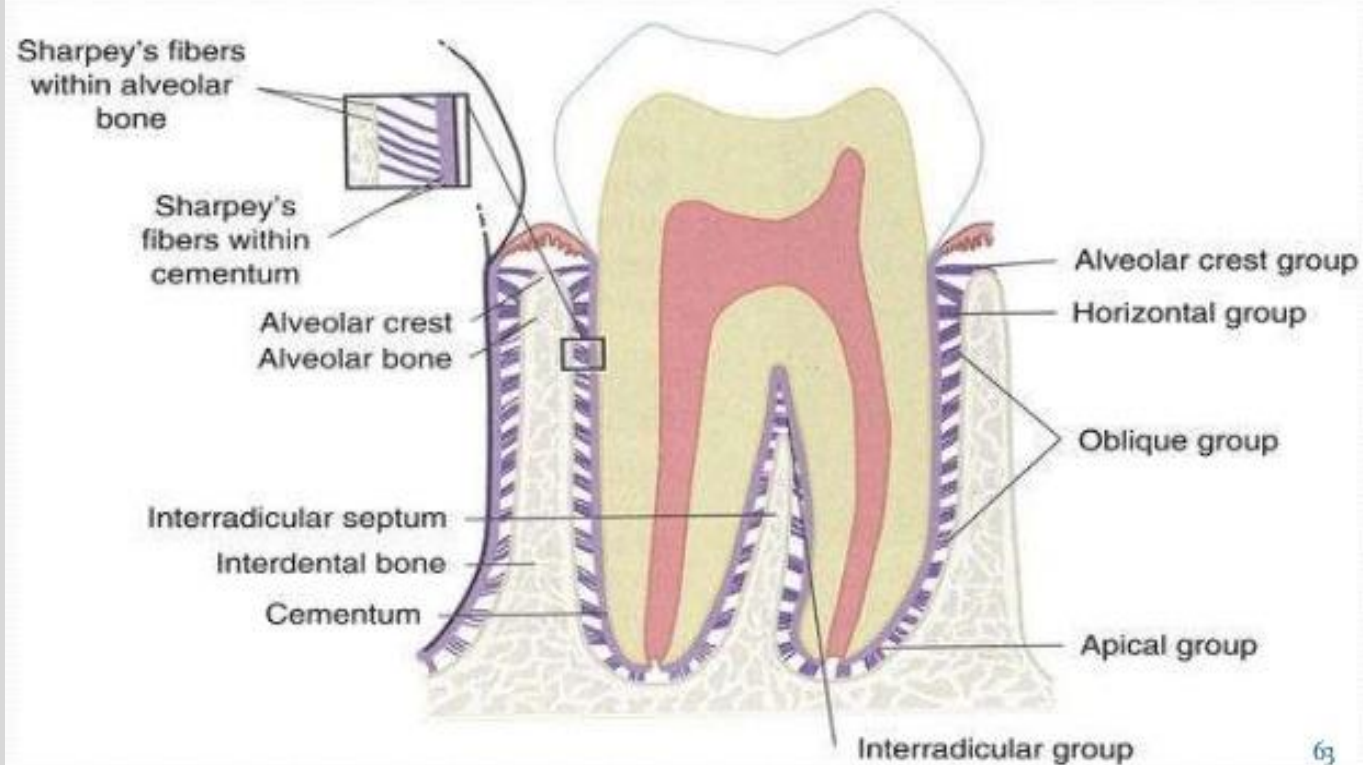


PRACTICAL-PERIODONTAL LIGAMENTS

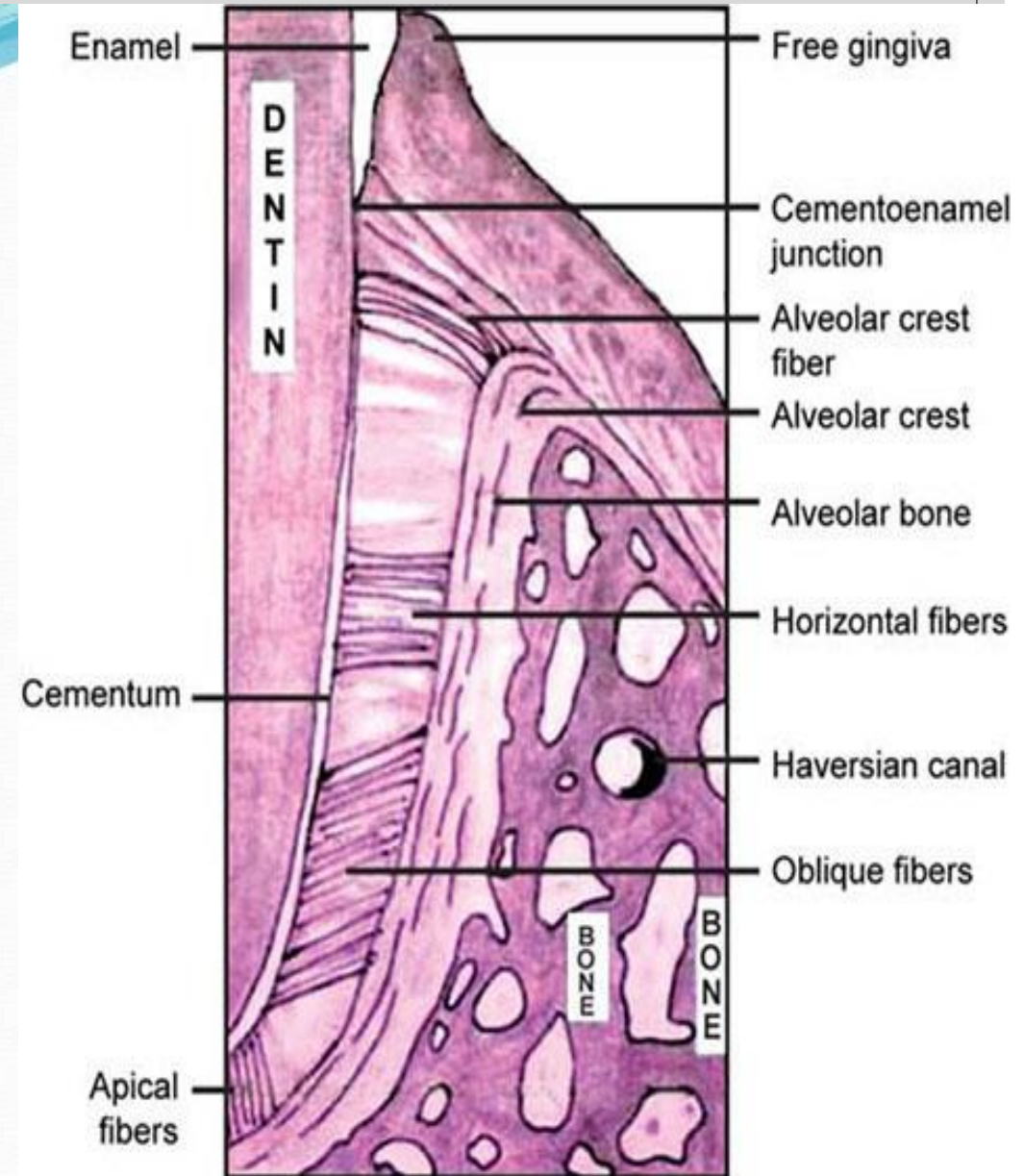
DR SAJDA GAJDHAR



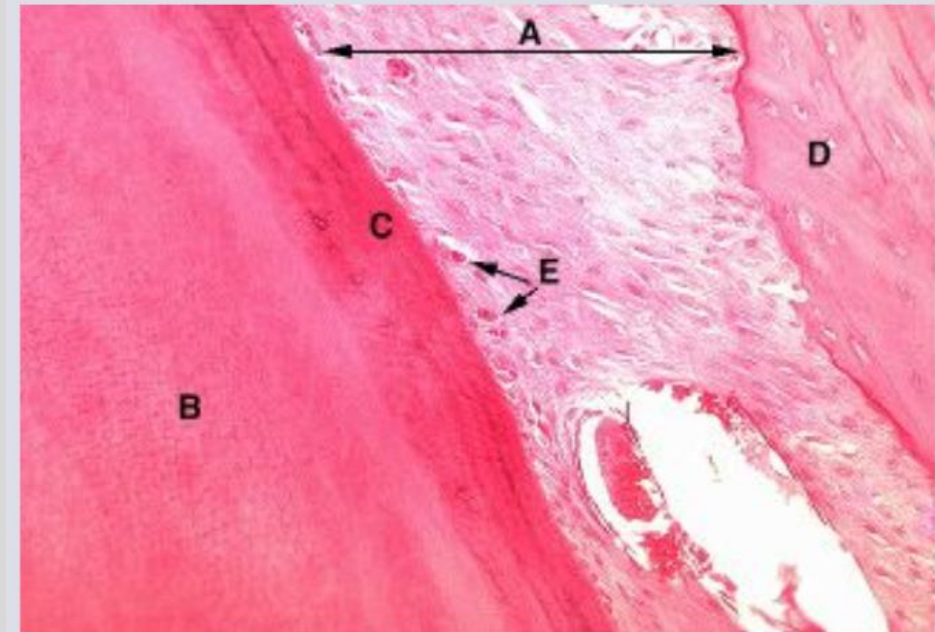
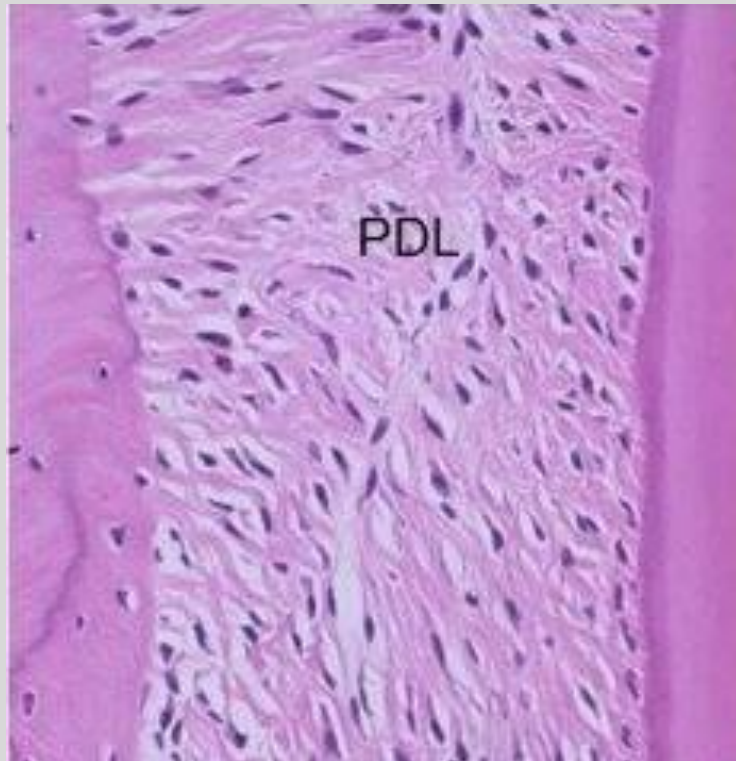
PERIODONTAL LIGAMENT FIBERS



63

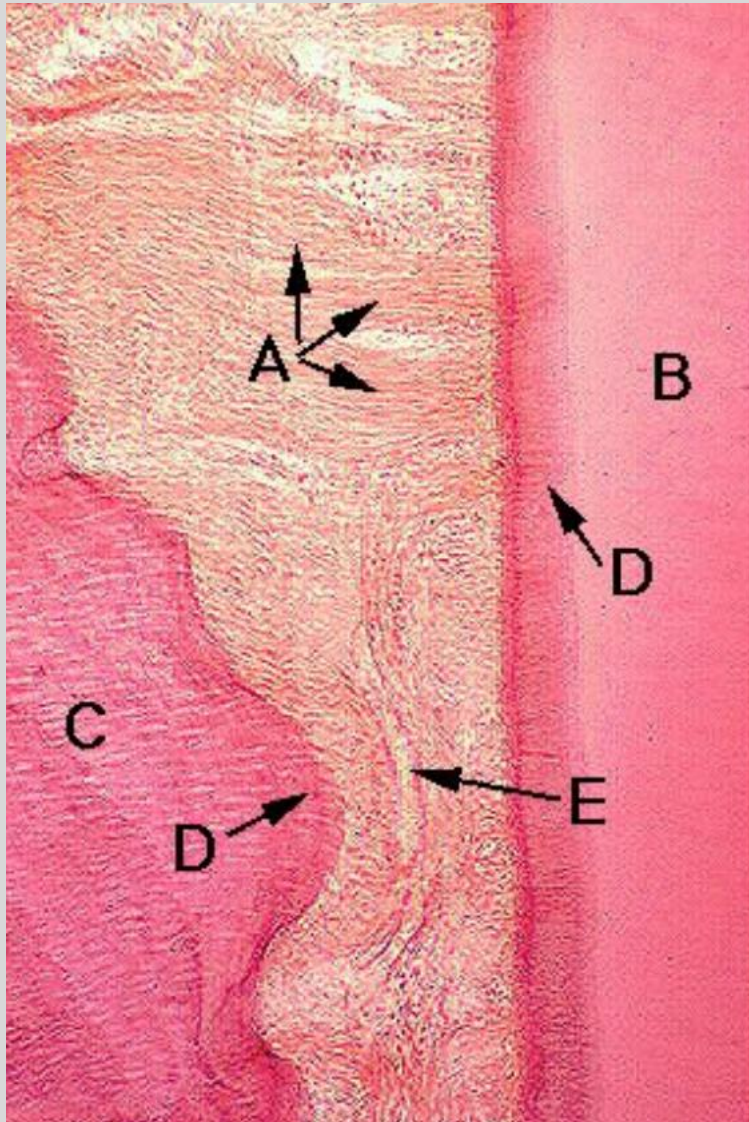


A number of cell types are found within the PDL (A), the most predominant being fibroblasts and macrophages. The principal cell type is the fibroblast and the principal intercellular substance is collagen.



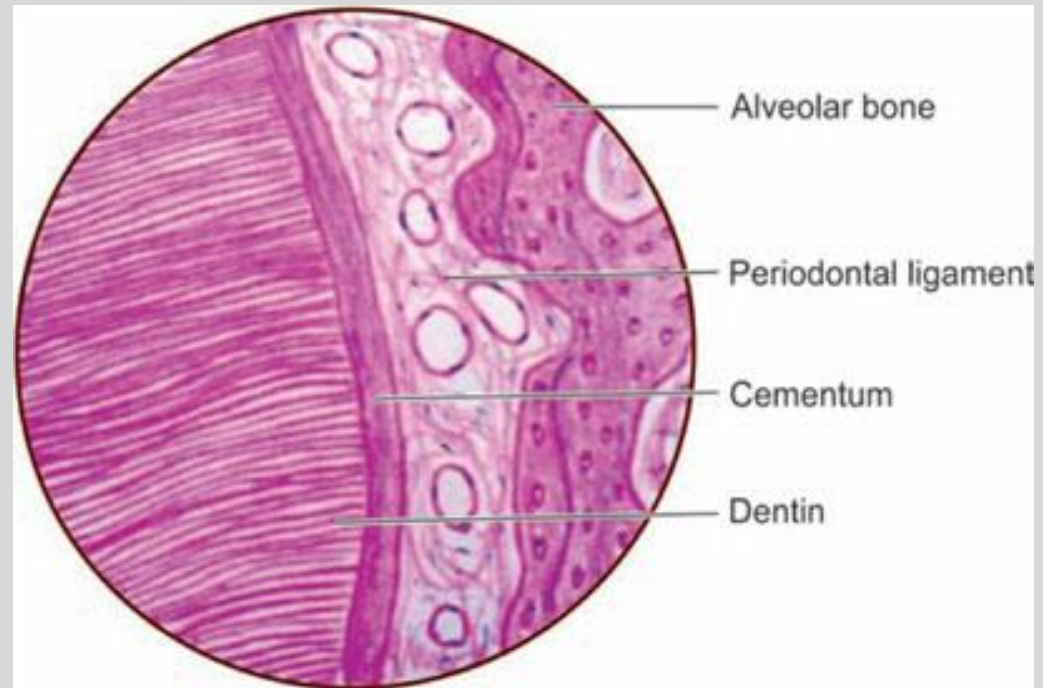
Legend

- | | |
|--------------------------|----------------------|
| A - periodontal ligament | D - alveolar bone |
| B - dentin | E - epithelial rests |
| C - cementum | |



A - horizontal principal
fibers
B - cementum
C - alveolar bone

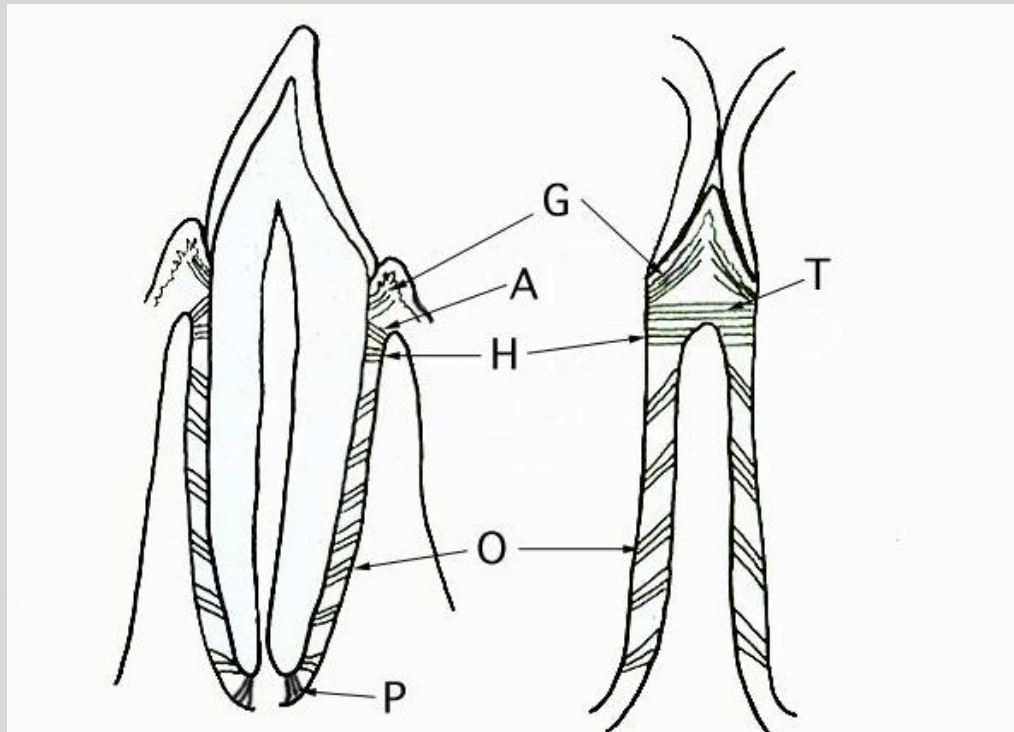
D - Sharpey's
fibers
E - blood vessel





PRINCIPAL FIBER GROUPS

Groups of collagen fibers arranged within the periodontal ligament, anchoring the tooth to bone. These groups of fibers are referred to as principal fibers



A - alveolar crest fibers

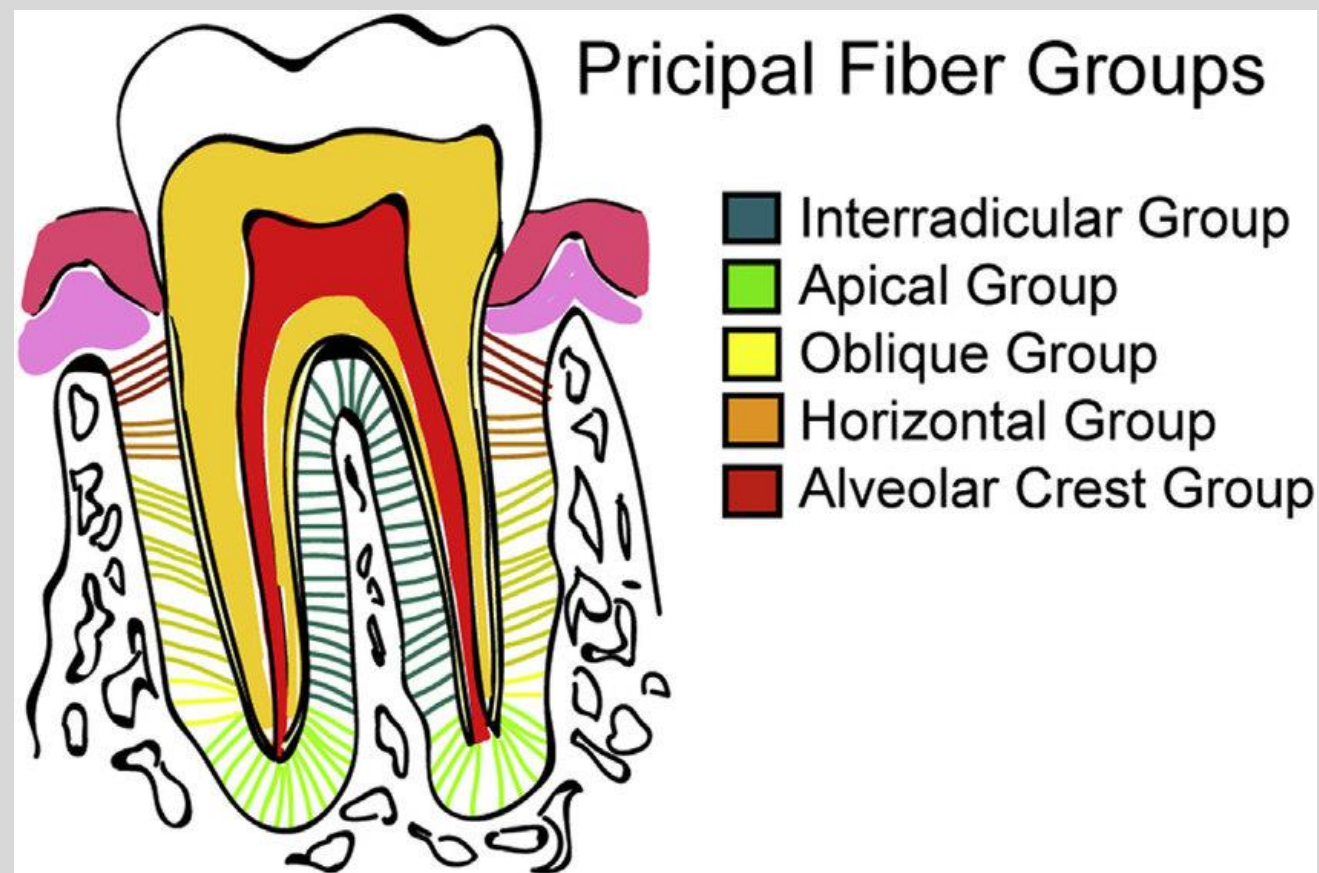
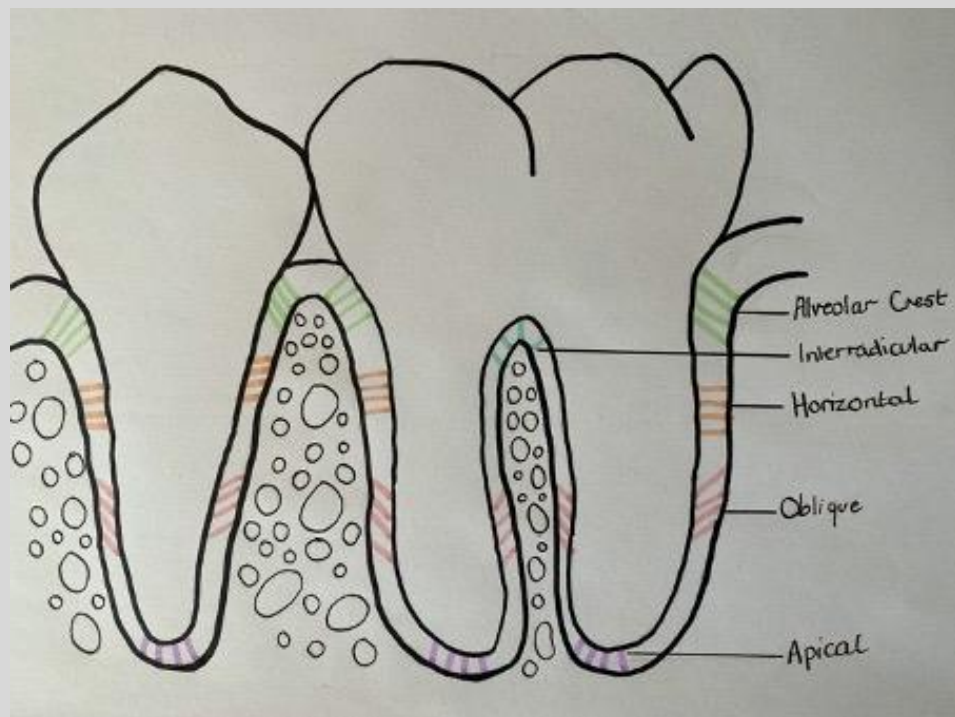
G - gingival fibers

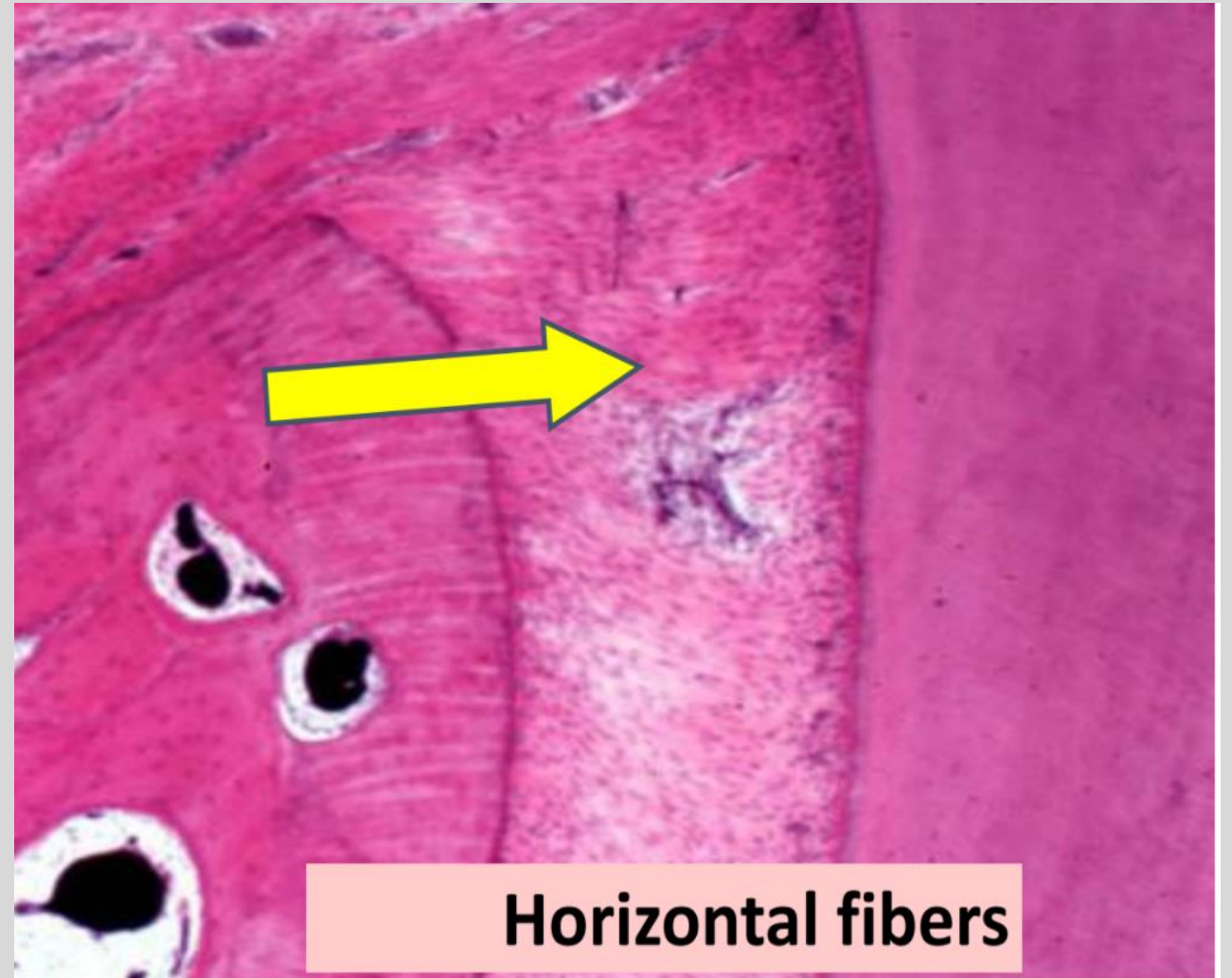
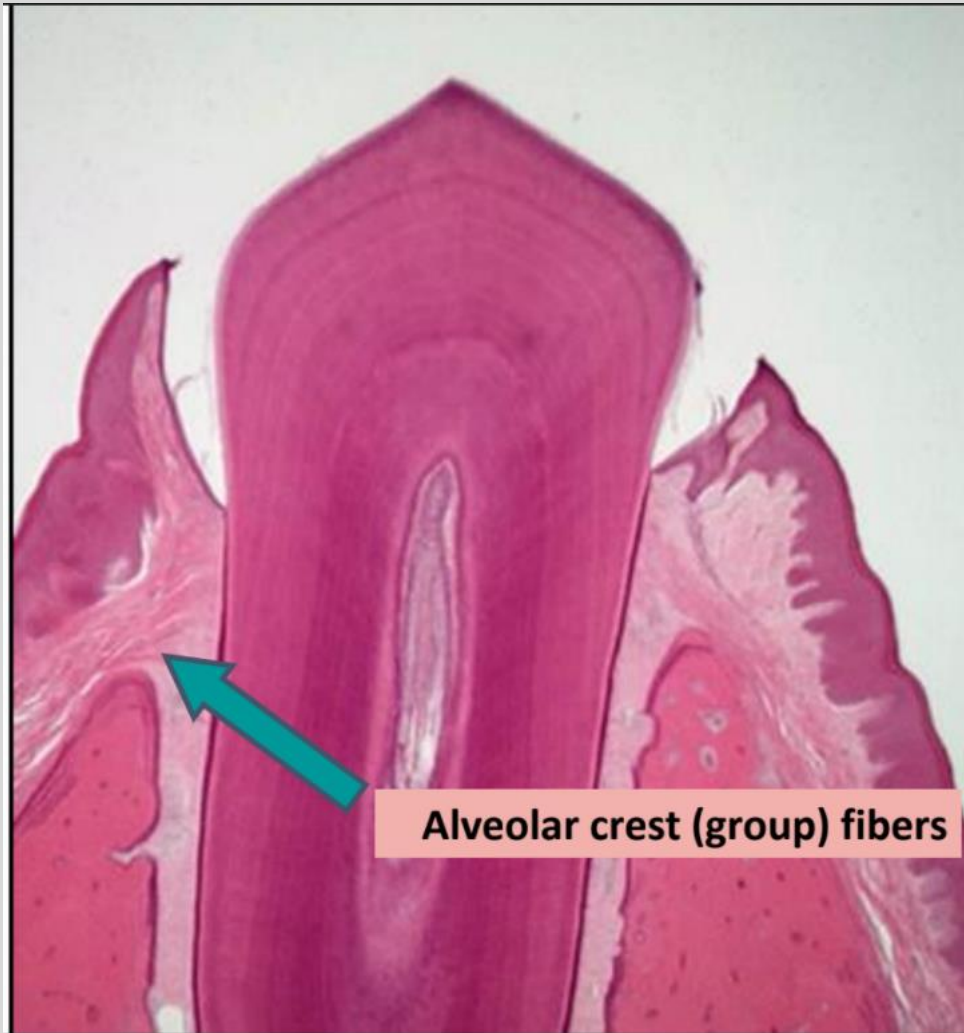
H - horizontal fibers

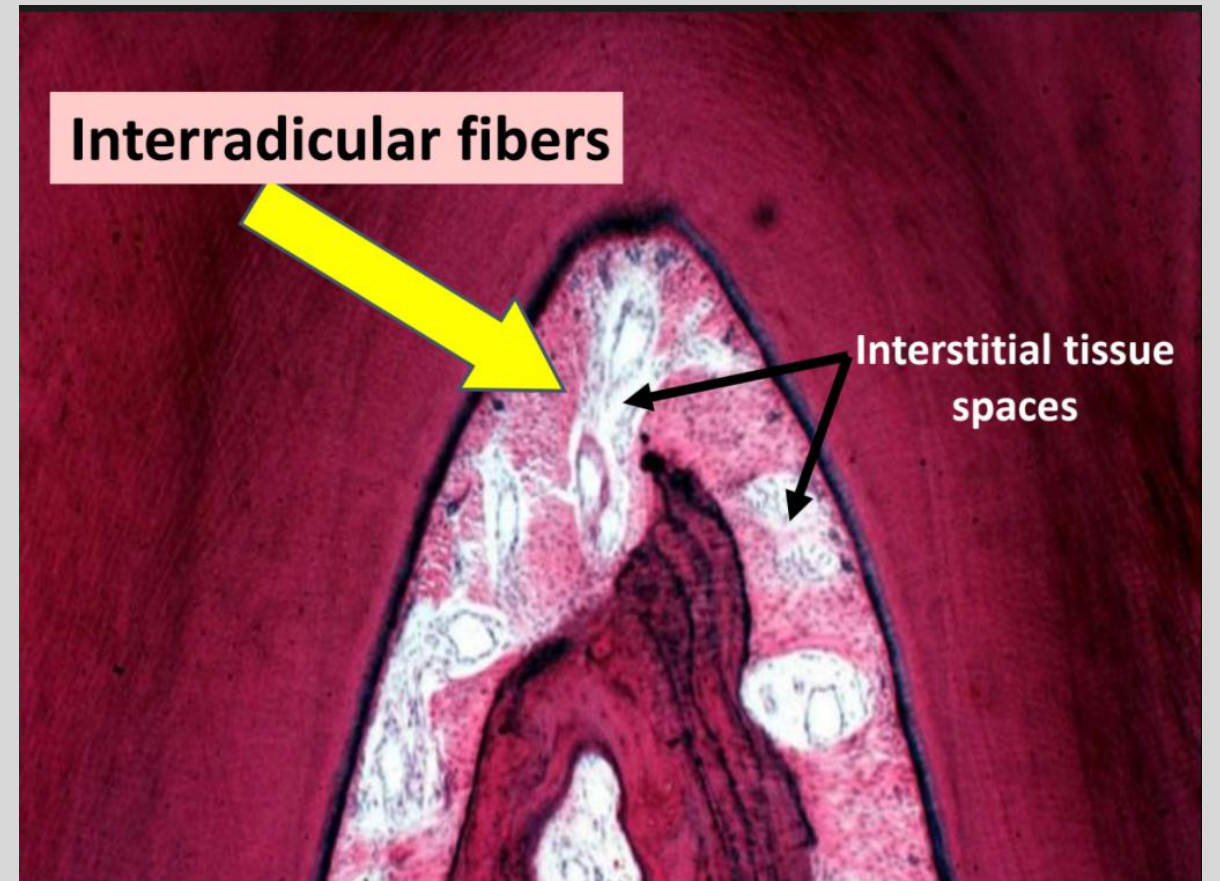
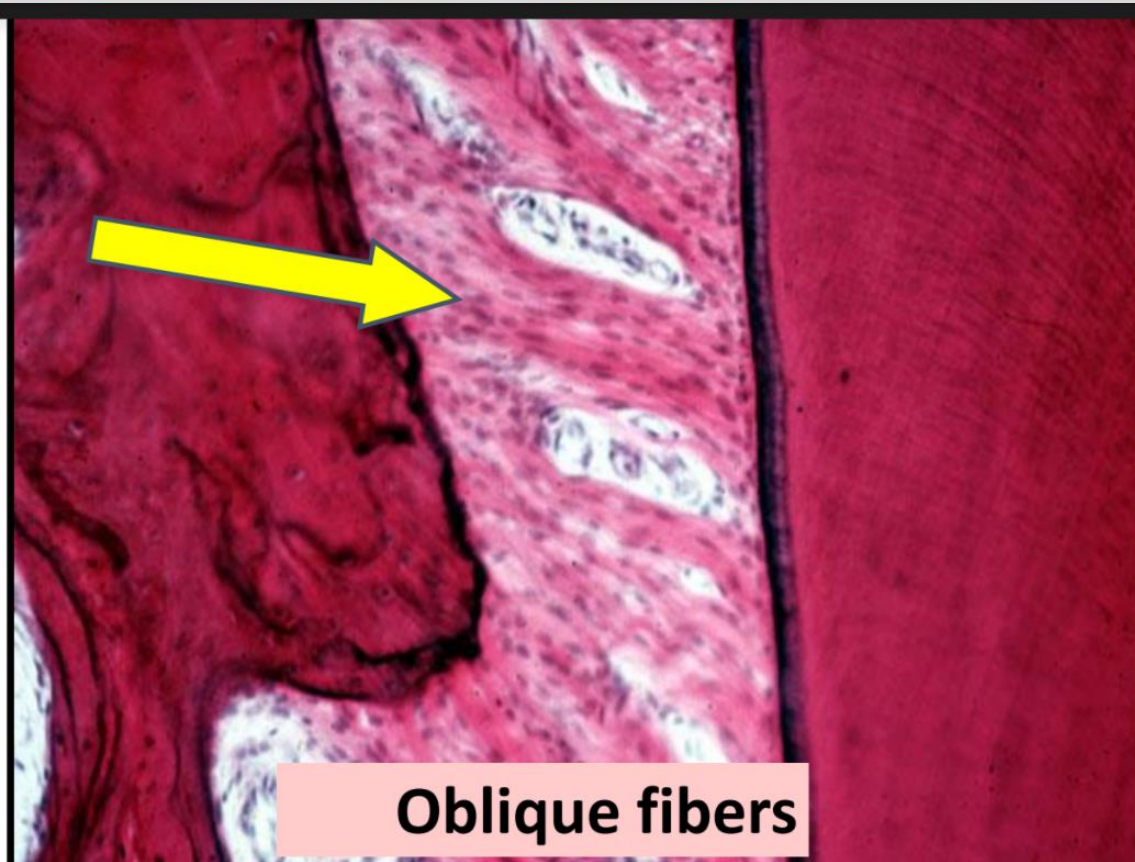
O - oblique fibers

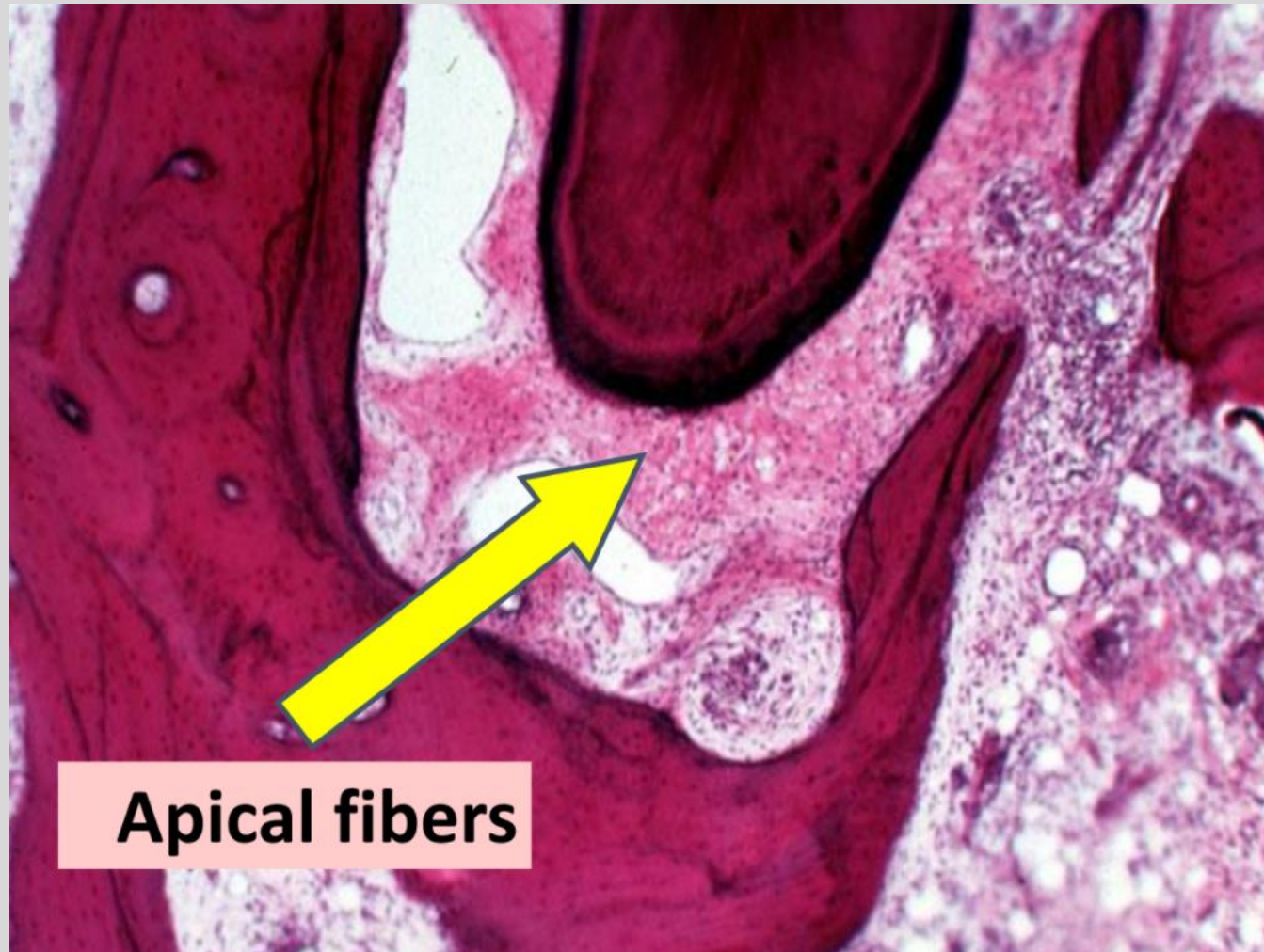
P - (peri)apical fibers

T - transseptal fibers

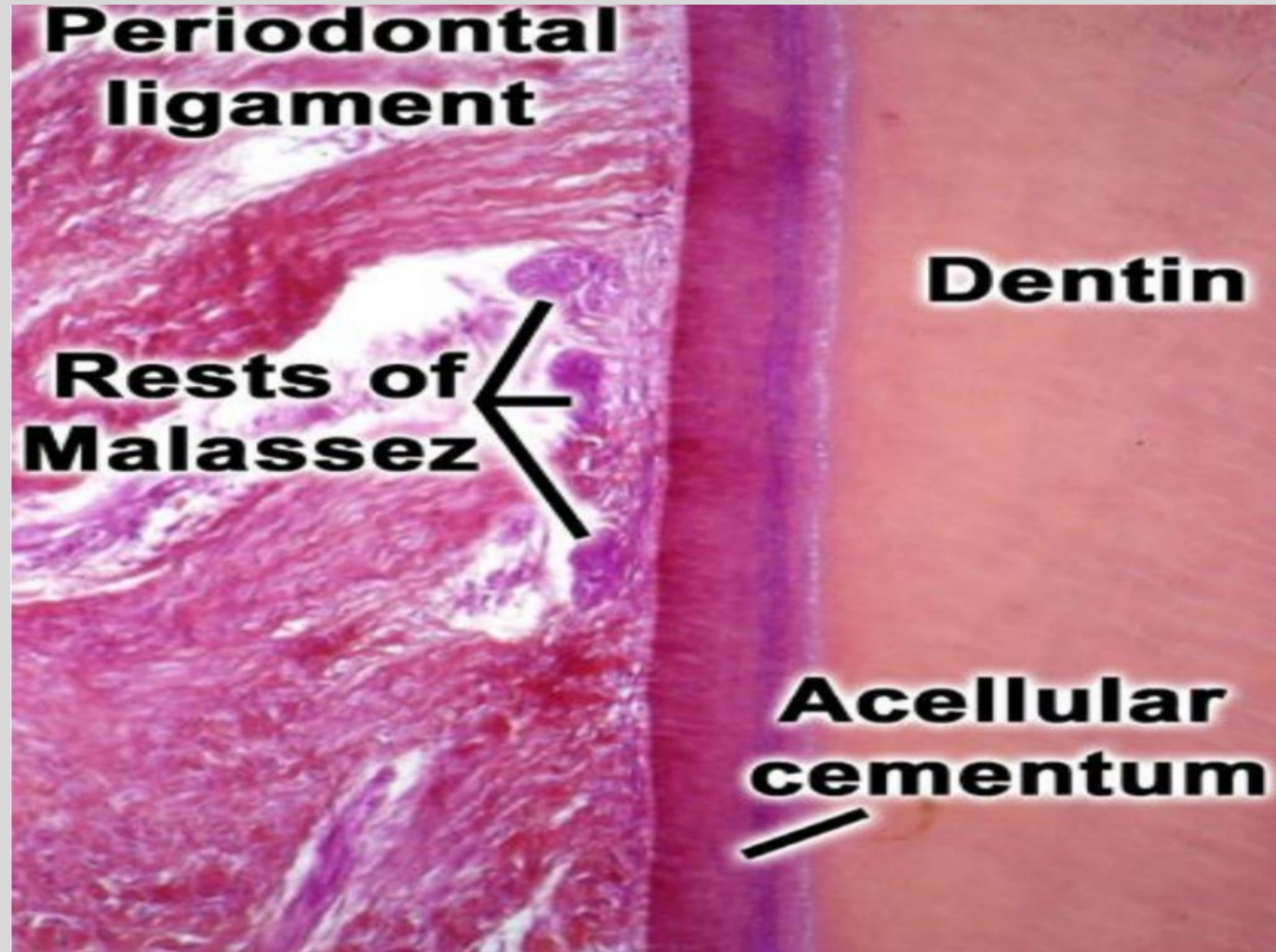




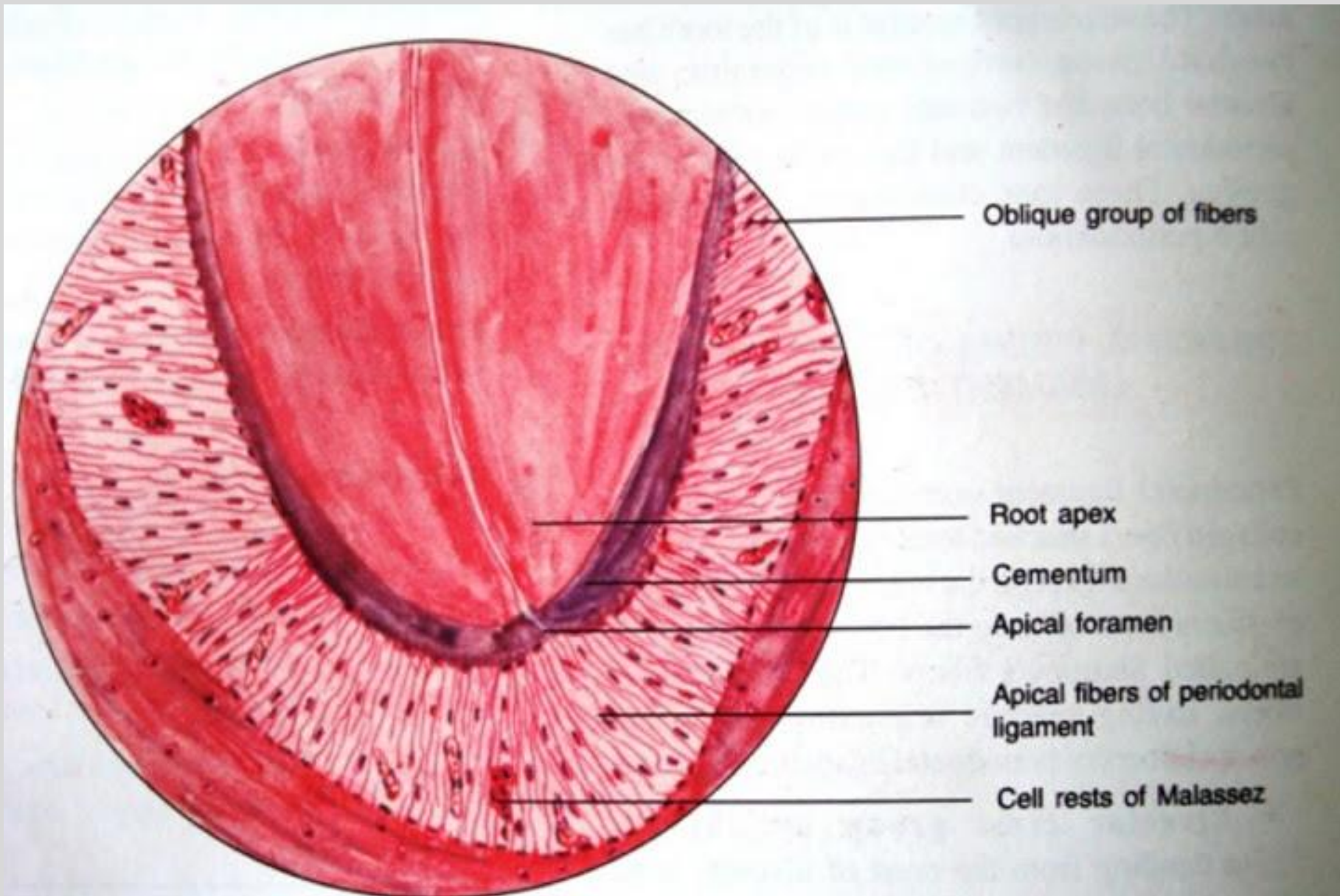




Apical fibers

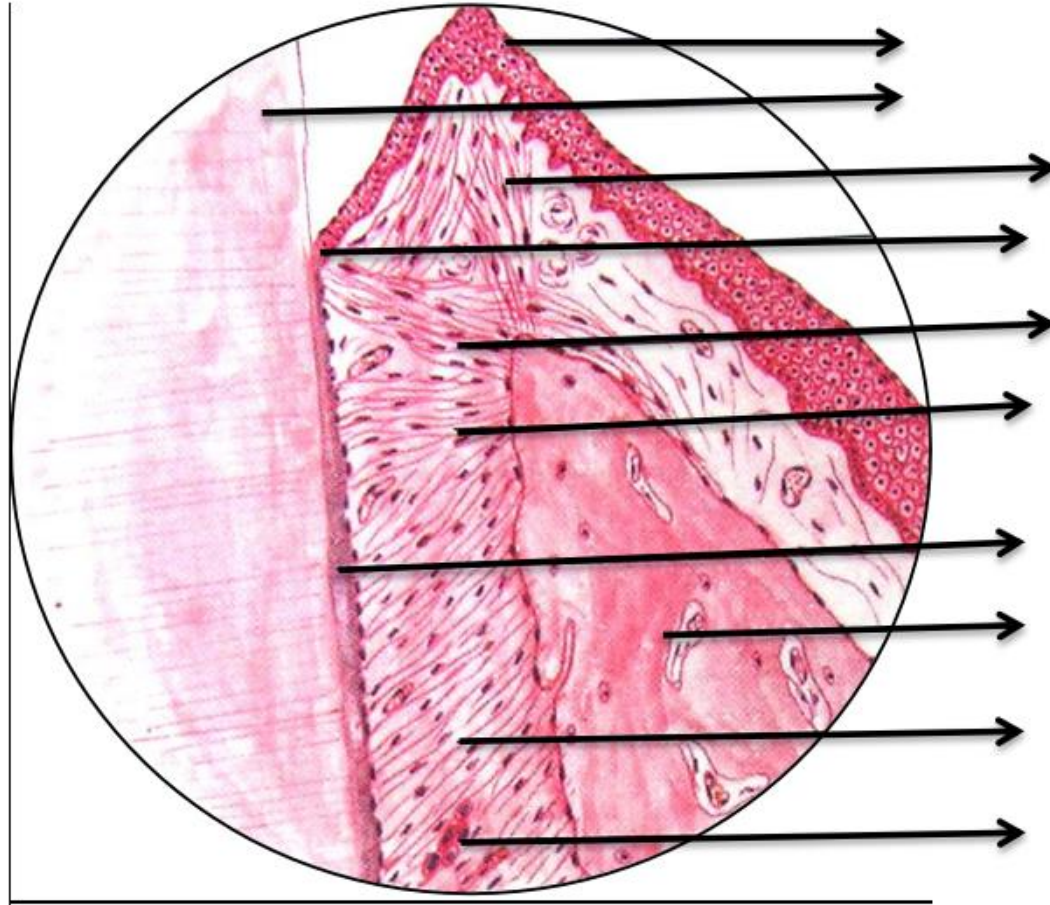


- Homogenous pink tooth like structure.....**DENTIN**
- Pink region containing multiple spaces.....**ALVEOLAR BONE**
- Fibrous connective tissue found between the two pink structures.....**Periodontal Ligament (dentoalveolar ligament)**
- A lot of spaces are found between the fibers.....(**Interstitial tissue spaces**)
- The fibers of the connective tissue are **oriented** in varies directions:
 - A- From the tooth structure to the gingiva (**GINGIVAL LIGAMENT**)
 - B- From the alveolar bone crest to the tooth neck..... (**ALV. CREST GROUP**)
 - C- Horizontally directed from the alv. bone to the cementum of the tooth.....(**HORIZONTAL GROUP**)
 - D- From the bone to the tooth with the higher attachment into the bone and lower attachment in the tooth and occupy most of the root length(**OBLIQUE GROUP**)
 - E- From the apex of the tooth radiating to the alveolar bone (**APICAL GROUP**)
 - F- From the inter-radicular bone radiating to the furcation area of the root (**INTERRADICULAR GROUP**)

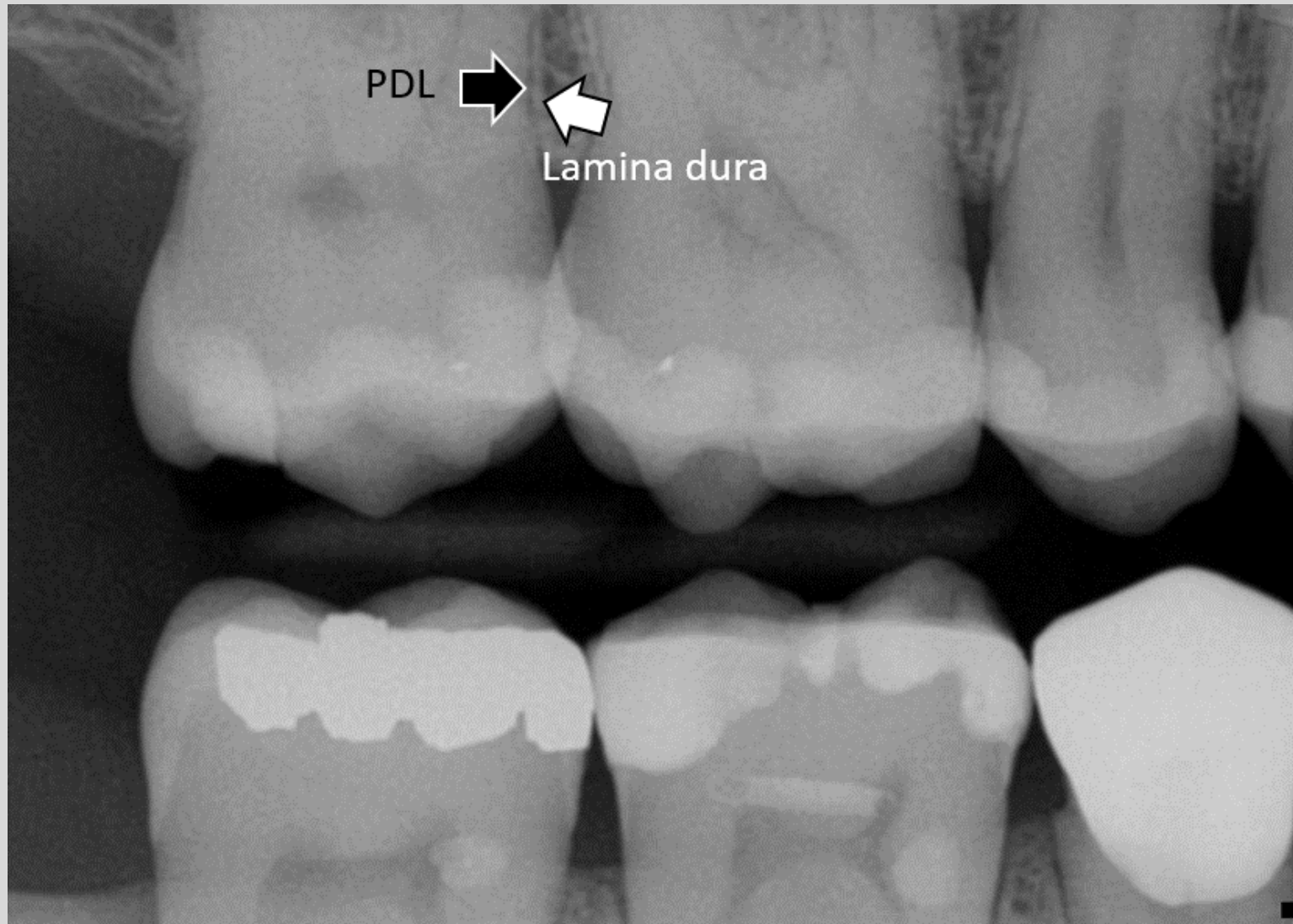


Exercise -12

Slide labelling:



Key features:



THANK YOU

